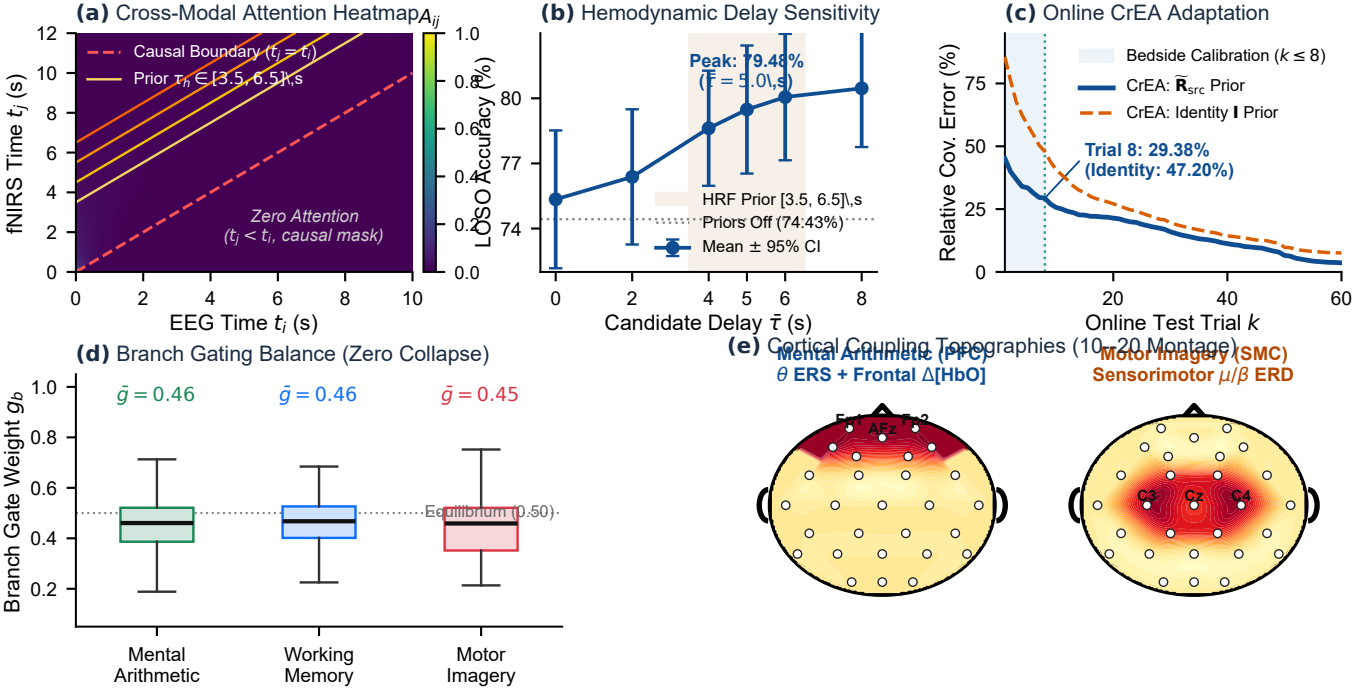

(a) Cross-Modal Attention Heatmap A_{ij}
Causal Boundary ($t_j = t_i$)
Prior $\tau_h \in [3.5, 6.5]$\,s
Zero Attention
($t_j < t_i$, causal mask)
fNIRS Time t_j (s)
EEG Time t_i (s)
LOSO Accuracy (%)
(b) Hemodynamic Delay Sensitivity
Peak: 79.48%
($\bar{\tau}$ = 5.0\,s)
HRF Prior [3.5, 6.5]\,s
Priors Off (74.43%)
Mean ± 95% CI
Candidate Delay $\bar{\tau}$ (s)
(c) Online CrEA Adaptation
Bedside Calibration ($k \leq 8$)
CrEA: $\widetilde{\mathbf{R}}_{\mathrm{src}}$ Prior
CrEA: Identity $\mathbf{I}$ Prior
Trial 8: 29.38%
(Identity: 47.20%)
Relative Cov. Error (%)
Online Test Trial k
(d) Branch Gating Balance (Zero Collapse)
$\bar{g}$ = 0.46
$\bar{g}$ = 0.46
$\bar{g}$ = 0.45
Equilibrium (0.50)
Branch Gate Weight g_b
Mental Arithmetic
Working Memory
Motor Imagery
(e) Cortical Coupling Topographies (10–20 Montage)
Mental Arithmetic (PFC)
θ ERS + Frontal Δ[HbO]
Motor Imagery (SMC)
Sensorimotor μ/β ERD
AFz
C3
Cz
C4